2024 SSMO Tiebreaker Round Solutions

SMO Team

Tiebreaker Round Problem 1: Compute the exact value of

$$2^2 + 0^2 + 2^2 + 4^2 + 20^2 + 22^2 + 24^2 + 40^2 + 42^2 + 202^2.$$

Answer: $\boxed{45652}$

Solution: We have

$$\begin{aligned} &2^2 + 0^2 + 2^2 + 4^2 + 20^2 + 22^2 + 24^2 + 40^2 + 42^2 + 202^2 \\ &= 4 + 0 + 4 + 16 + 400 + 484 + 576 + 1600 + 1764 + 40804 = \boxed{45652}. \end{aligned}$$

Tiebreaker Round Problem 2: Bob is attempting to make a 3-point shot. Bob attempts the shot 97 times. Each time, Bob has a 35% chance of making the shot. If S_1 denotes Bob's expected score and S_2 the score Bob is most likely to get, then $|S_1 - S_2|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{23}$

Solution: Firstly, the expected number of shots Bob makes is $\left(\frac{35}{100}\right) \cdot 97$. Now, the number of shots Bob is most likely to make is the largest integer a such that

$$\begin{aligned} \left(\frac{35}{100}\right)^a \left(\frac{65}{100}\right)^{97-a} \binom{97}{a} \geq \left(\frac{35}{100}\right)^{a-1} \left(\frac{65}{100}\right)^{98-a} \binom{97}{a-1} &\implies \\ 35(98-a) \geq 65(a) &\implies \\ 3430 \geq 100a &\implies \\ 34.3 \geq a. \end{aligned}$$

So, Bob is most likely to make 34 shots. Therefore, the answer is

$$\left|3\left(\frac{35 \cdot 97}{100} - 34\right)\right| = \frac{3}{20} \implies 3 + 20 = \boxed{23}.$$

Tiebreaker Round Problem 3: Let $A = \ldots a_2 a_1 a_0 . a_{-1} a_{-2} a_{-3} \ldots$ be a positive terminating decimal. The length of A is defined to be the length of the shortest sub-sequence of consecutive digits that includes all nonzero digits and at least one of a_0, a_{-1}. So, the length of 12.03 is 4 and the length of 0.123 is 3. Let $f(n)$ be the average of all positive terminating decimals of length n. Find the value of $\left\lfloor \sum_{n=1}^{10} (n+1) f(n) \right\rfloor$.

Answer: $\boxed{7319584052}$

Solution: Consider a number of length n, and suppose its shortest sub-sequence runs from a_h down to a_{h-n+1}. Since this sub-sequence contains a_0 or a_{-1}, we have $-1 \le h \le n-1$. Note that a_h must be nonzero unless $h = -1$, and a_{h-n+1} must be nonzero unless $h - n + 1 = 0$, since otherwise a shorter sub-sequence would work. Every other digit of the sub-sequence can be any digit. We will do casework on h, using that the average nonzero digit is 5 and the average digit is $\frac{9}{2}$. If $h = -1$, there are $9 \cdot 10^{n-1}$ such numbers, with average

$$\frac{9}{2} \sum_{k=-n}^{-1} 10^k + \frac{1}{2} \cdot 10^{-n} = \frac{1}{2}.$$

If $h = n - 1$, there are $9 \cdot 10^{n-1}$ such numbers, with average

$$5 \cdot 10^{n-1} + \frac{9}{2} \sum_{k=0}^{n-2} 10^k = \frac{11 \cdot 10^{n-1} - 1}{2}.$$

Finally, if $0 \le h \le n - 2$, there are $81 \cdot 10^{n-2}$ such numbers, with average

$$5 \cdot 10^h + 5 \cdot 10^{h-n+1} + \frac{9}{2} \sum_{k=h-n+2}^{h-1} 10^k = \frac{11}{2} \cdot 10^h.$$

So, there are

$$2 \cdot 9 \cdot 10^{n-1} + (n-1) \cdot 81 \cdot 10^{n-2} = 9 \cdot 10^{n-2}(9n + 11)$$

numbers of length n, and their sum is

$$\begin{aligned} &9 \cdot 10^{n-1} \cdot \frac{1}{2} + 9 \cdot 10^{n-1} \cdot \frac{11 \cdot 10^{n-1} - 1}{2} + 81 \cdot 10^{n-2} \cdot \frac{11}{2} \sum_{h=0}^{n-2} 10^h \\ &= \frac{99}{2} \cdot 10^{2n-2} + \frac{99}{2} \cdot 10^{n-2} \left(10^{n-1} - 1\right) \\ &= \frac{99}{2} \cdot 10^{n-2} \left(11 \cdot 10^{n-1} - 1\right). \end{aligned}$$

Dividing, we have

$$(n+1)f(n) = \frac{11(n+1)\left(11 \cdot 10^{n-1} - 1\right)}{2(9n + 11)}.$$

Computing each term to three decimal places, we have

n	$(n+1)f(n)$
1	5.500
2	62.017
3	636.263
4	6435.585
5	64820.839
6	651537.869
7	6540539.946
8	65602409.042
9	657608695.054
10	6589108910.292

Since the total rounding error is less than 0.005, the sum is 7319584052.407 to within 0.005, giving an answer of $\boxed{7319584052}$.